The center $Z(Q_8)$ is $\{\pm 1\}$, and $Q_8/Z(Q_8) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. The four 1-dimensional irreducible representations of $\mathbb{Z}_2 \times \mathbb{Z}_2$ can be "pulled back" to Q_8. That is, if $q : Q_8 \to Q_8/Z(Q_8)$ is the quotient map and ρ is any representation of $Q_8/Z(Q_8)$, then $\rho \circ q$ gives a representation of Q_8.

The 2-dimensional representation is $V = \mathbb{C}^2$, given by $\rho(-1) = -\mathrm{Id}$ and

$$\begin{aligned} \rho(i) &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \\ \rho(j) &= \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix}, \\ \rho(k) &= \begin{pmatrix} 0 & -\sqrt{-1} \\ -\sqrt{-1} & 0 \end{pmatrix}. \end{aligned} \tag{4.3.1}$$

These are the Pauli matrices, which arise in quantum mechanics.

Exercise 4.3.1. Show that the 2-dimensional irreducible representation of Q_8 can be realized in the space of functions $f : Q_8 \to \mathbb{C}$ such that $f(gi) = \sqrt{-1} f(g)$ (the action of G is by right multiplication, $g \circ f(x) = f(xg)$).

(4) The symmetric group S_4. The order of S_4 is 24, and there are five conjugacy classes: $e, (12), (123), (1234), (12)(34)$. Thus the sum of the squares of the dimensions of five irreducible representations is 24. As with S_3, there are two of dimension 1: the trivial and sign representations, $\mathbb{C}_+$ and $\mathbb{C}_-$. The other three must then have dimensions 2, 3, and 3. Because $S_3 \cong S_4/\mathbb{Z}_2 \times \mathbb{Z}_2$, where $\mathbb{Z}_2 \times \mathbb{Z}_2$ is $\{e, (12)(34), (13)(24), (14)(23)\}$, the 2-dimensional representation of S_3 can be pulled back to the 2-dimensional representation of S_4, which we will call $\mathbb{C}^2$.

We can consider S_4 as the group of rotations of a cube acting by permuting the interior diagonals (or, equivalently, on a regular octahedron permuting pairs of opposite faces); this gives the 3-dimensional representation $\mathbb{C}^3_+$.

The last 3-dimensional representation is $\mathbb{C}^3_-$, the product of $\mathbb{C}^3_+$ with the sign representation. $\mathbb{C}^3_+$ and $\mathbb{C}^3_-$ are different, for if g is a transposition, $\det g|_{\mathbb{C}^3_+} = 1$ while $\det g|_{\mathbb{C}^3_-} = (-1)^3 = -1$. Note